

APM1205 – Applied Regression Analysis

Formative Assessment 4

Dummy-Variable Regression Using the *diamonds* Dataset

Total: 40 points

This report completes Parts A–E using the actual diamonds dataset supplied with ggplot2. Ideal is used as the reference category, following the assessment example. The significance level is $\alpha = 0.05$.

Part A. Exploring and Preparing the Data

A1. Observations and variables. The dataset contains **53,940 observations** and **10 variables**.

A2. Response variable. The response variable is **price**, which measures the price of a diamond in US dollars.

A3. Quantitative explanatory variable. The quantitative explanatory variable is **carat**, which measures diamond weight.

A4. Categorical explanatory variable. The categorical explanatory variable is **cut**, with five categories: Fair, Good, Very Good, Premium, and Ideal.

A5. Reference category. The selected reference category is **Ideal**. A reference category is needed because a five-level categorical variable requires four dummy variables when an intercept is included. Using all five dummies together with an intercept would create perfect multicollinearity.

R commands:

```
library(ggplot2) data(diamonds) head(diamonds) str(diamonds) summary(diamonds)
```

Part B. Constructing Dummy Variables

B1. Dummy-variable coding. With Ideal as the reference category, the coding is:

Cut	D1 Fair	D2 Good	D3 Very Good	D4 Premium
Fair	1	0	0	0
Good	0	1	0	0
Very Good	0	0	1	0
Premium	0	0	0	1
Ideal	0	0	0	0

B2. All-zero category. The all-zero category is **Ideal**. An Ideal diamond has all four dummy variables equal to zero, so the intercept and carat coefficient describe the reference group before dummy-variable adjustments.

B3. Why not five dummies with an intercept? For every observation, the five dummy variables sum to one. Since the intercept is also a column of ones, one dummy can be written exactly as a combination of the intercept and the other four dummies. This is perfect multicollinearity, so four dummies plus a reference category are used.

Part C. Additive Dummy-Variable Regression

The fitted additive model is $\text{Price} = \beta_0 + \beta_1(\text{Carat}) + \text{cut effects} + \varepsilon$, where cut effects are measured relative to Ideal.

C1. Estimated regression equation.

$\text{Price} = -2074.546 + 7871.082(\text{Carat}) - 1800.924(\text{Fair}) - 680.592(\text{Good}) - 290.789(\text{Very Good}) - 361.847(\text{Premium})$

C2. Carat coefficient. The estimated carat coefficient is **7871.082**. Holding cut constant, a one-carat increase is associated with an estimated increase of about **\$7,871.082** in price under the additive model.

C3. Dummy coefficient. The Premium coefficient is **-361.847**. Holding carat constant, a Premium-cut diamond has an estimated price about **\$361.847 lower** than an Ideal-cut diamond.

C4. R². The additive model has $R^2 = 0.856$. Thus approximately **85.6%** of observed price variation is explained by carat and cut in this additive linear model.

C5. Slopes. The additive model gives all cut categories the **same carat slope**. The dummy variables change the intercepts only, so the fitted lines are parallel.

R commands:

```
model1 <- lm(price ~ carat + cut, data = diamonds) summary(model1)
```

Part D. Interaction Between Carat and Cut

D1. Hypotheses. H0: all four carat × cut interaction coefficients are zero; the carat effect is the same across cut categories. HA: at least one interaction coefficient is nonzero; the carat effect differs for at least one cut category.

D2. Interaction coefficients and p-values.

Interaction	Estimate	p-value
Fair × Carat	-2267.896	< 0.001
Good × Carat	-712.755	< 0.001
Very Good × Carat	-256.419	< 0.001
Premium × Carat	-384.639	< 0.001

D3. Incremental F-test. Comparing the nested additive and interaction models gives $F = 247.735$ with 4 and 53930 degrees of freedom and $p < 0.001$.

D4. Decision at $\alpha = 0.05$. Because $p < 0.05$, reject H0. The data provide statistical evidence that the effect of carat on price differs across cut categories. The interaction model has $R^2 = 0.859$, compared with 0.856 for the additive model.

D5. Practical difference. The additive model permits different intercepts but a common carat slope. The interaction model permits the carat slope to vary across cut categories, so the fitted lines need not be parallel.

R commands:

```
model2 <- lm(price ~ carat * cut, data = diamonds) summary(model2) anova(model1, model2)
```

Part E. Visualization and Statistical Conclusion

E1. Required visualization. The scatterplot below shows Price versus Carat by cut, with separate fitted regression lines corresponding to the interaction model.



E2. Statistical conclusion. Diamond price has a strong positive linear association with carat: heavier diamonds tend to have higher prices. Cut also contributes to the regression model, with the dummy variables representing price differences between each cut category and the Ideal reference category while holding carat constant. In the additive model, all cut categories share a common carat slope, but the interaction analysis provides strong statistical evidence that this common-slope assumption is not adequate. The incremental F-test comparing the additive and interaction models gives $F = 247.735$ with $p < 0.001$, so the interaction terms improve the model at $\alpha = 0.05$. The interaction model therefore allows the estimated effect of carat on price to vary across cut categories. Overall, the regression analysis indicates that both diamond weight and cut are associated with price, and their relationship is better represented when the carat effect is allowed to differ by cut.

Appendix. Core Reproducible R Code

The submitted FA4_Dummy_Regression.R file contains the complete standalone R code. The core commands are reproduced here for reference.

```
library(ggplot2) data(diamonds) diamonds$cut <- relevel(diamonds$cut, ref = "Ideal") diamonds$D1_Fair <-  
as.integer(diamonds$cut == "Fair") diamonds$D2_Good <- as.integer(diamonds$cut == "Good")  
diamonds$D3_VeryGood <- as.integer(diamonds$cut == "Very Good") diamonds$D4_Premium <-  
as.integer(diamonds$cut == "Premium") model1 <- lm(price ~ carat + cut, data = diamonds) model2 <-  
lm(price ~ carat * cut, data = diamonds) summary(model1) summary(model2) anova(model1, model2) p <-  
ggplot(diamonds, aes(carat, price, color = cut)) + geom_point(alpha = 0.3) + geom_smooth(method = "lm",  
se = FALSE) + labs(title = "Diamond Price versus Carat by Cut", x = "Carat", y = "Price (US Dollars)")  
ggsave("figures/diamond_price_vs_carat_by_cut.png", p, width = 9, height = 6, dpi = 300)
```